First-Order Initial Value Problems (IVPs)

1.] Let $u(t)$ be the quantity or concentration of a substance at time t. Then $u(t)$ experiences exponential growth or decay if it satisfies the following initial value problem:

$$\frac{du}{dt} = ku \qquad u(0) = u_0,$$

where u_0 is known data. The parameter k is the per capita growth/decay rate. Solve this IVP.

2.] According to Newton's Law of Cooling, the rate of change of temperature of a cooling object is proportional to the temperature *difference* between the object and the surrounding medium. That is,

$$\frac{du}{dt} = -k\left(u - u_s\right), \qquad u(0) = u_0$$

where $u(t)$ is the temperature of the object at time t, and u_s is the temperature of the surroundings, assumed to be constant. Here, $k > 0$. Solve this IVP. What is the equilibrium temperature?

3.] Let $u(t)$ be the size of a population of interest at time t. Given limited resources or space to expand, many populations experience *logistic* growth at a per capita rate $k > 0$ to a *carrying capacity*, m:

$$\frac{du}{dt} = ku\left(1 - \frac{u}{M}\right), \qquad y(0) = y_0.$$

Here, k and M are both strictly positive. Show that $u(t) = \dfrac{Mu_0}{u_0 + (M - u_0)e^{-kt}}$ is a solution. Solve this equation numerically in MATLAB and show several solutions.

4.] Create an m-file that computes the numerical solution to the following initial value problem

$$\frac{du}{dt} = -ku\left(1 - \frac{u}{m}\right)\left(1 - \frac{u}{M}\right) \qquad u(0) = u_0,$$

for parameter values $k > 0$, $m > 0$, and $M > 0$ and an initial value y_0. Your file should plot the solution to the equation with many solutions on the same graph, with $u_0 < m$, $m < u_0 < M$, and one with $u_0 > M$.